Chapter 14 — Request Prioritization, Queueing & Client-Side Rate Limiting

> **Part II — Networking Patterns** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 1 (Networking Core — OkHttp `Dispatcher`), Ch 2 (reliability/backoff), Ch 11 (single-flight). Closes Part II's networking patterns. Applied in Ch 56 (image priority), Ch 58/59 (media), Ch 30 (background work).
>
> **Reference surface in StreamCart:** making the visible product grid's images and the current screen's data load before off-screen prefetch and analytics.

14.1 Problem Definition and Real-World Context

A mobile app issues many concurrent requests — UI-critical reads, images, prefetch, analytics, uploads — over **one** constrained, high-latency radio link. Treating them all equally hurts the experience:

- **Head-of-line blocking.** A burst of low-priority prefetch/analytics requests saturates the connection pool, so the data the user is *waiting on* queues behind work they don't care about.
- **Wasted bandwidth on stale work.** The user scrolls past a row, but its image requests still run; they should be **cancelled/deprioritized**.
- **Self-inflicted rate limiting.** Firing unbounded concurrent requests trips server 429s and wastes battery/radio.
- **Thundering analytics/telemetry.** Hundreds of small events sent individually thrash the radio (keeping it awake costs battery); they should be **batched**.

The solution is a small **scheduling policy** over the networking core: **prioritize** (user-visible > prefetch > background), **bound concurrency** (limit in-flight requests, per host), **queue** by priority, **cancel** work that no longer matters, and **rate-limit/batch** low-value traffic. These are policy decisions layered on Ch 1's client.

Three practical use cases

1. **E-commerce grid (StreamCart Shop).** Visible-cell images load first; off-screen prefetch yields; scrolled-away image requests are cancelled (Ch 56).
2. **OTT home (Watch).** The current row's artwork and the selected title's metadata outrank background prefetch of other rows.
3. **Analytics/telemetry (all).** Events are batched and flushed on a timer / size threshold / lifecycle, not sent one-by-one (Ch 82).

Why this is architectural

It defines **how a scarce shared resource (the network) is allocated** across competing needs — priority ordering, concurrency bounds, cancellation, and batching — a cross-cutting policy that shapes perceived performance and

battery.

14.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Assign priorities (e.g., Immediate / UserVisible / Prefetch / Background).
F2	Higher-priority requests start before lower ones when capacity is limited.
F3	Bound total and per-host in-flight concurrency.
F4	Cancel requests that are no longer needed (scrolled away, screen left).
F5	Rate-limit / throttle low-value traffic to avoid 429s and battery drain.
F6	Batch high-frequency small requests (analytics) instead of sending each.

Non-functional requirements

Attribute	Policy
Latency	User-visible requests not blocked by background work.
Battery	Fewer radio wakeups; batch; coalesce; respect Doze (Ch 47).
Network	Bounded concurrency; respect `Retry-After` /429 (Ch 2).
Reliability	Dropped/deprioritized work doesn't break correctness.
Fairness	Avoid starving low priority forever (aging, optional).

Scope

- **In scope:** priority model, bounded-concurrency scheduler/semaphore, OkHttp dispatcher tuning, cancellation on scroll/navigation, client rate limiting, analytics batching.
- **Out of scope:** retry/backoff (Ch 2), single-flight dedup (Ch 11), background deferrable work (WorkManager, Ch 30), image pipeline specifics (Ch 56).

Assumptions & constraints

- Coroutines; OkHttp `Dispatcher`; priorities are a small enum. Client rate limiting is **not** a security control (backend enforces, Ch 2). `minSdk 24`.

14.3 System Design Fundamentals

- **Priority classes.** A small ordered set: `Immediate` (blocking the user, e.g. tapped detail), `UserVisible` (on-screen content/images), `Prefetch` (likely-next, off-screen), `Background` (analytics, cleanup). Ordering, not

fine-grained numbers, is enough.

- **Bounded concurrency.** The radio and connection pool are finite; cap in-flight requests (a `Semaphore` or the OkHttp `Dispatcher`'s `maxRequests` / `maxRequestsPerHost`). Unbounded concurrency causes contention and 429s.
- **Priority scheduling.** When capacity is full, pick the highest-priority queued request next (a priority queue / per-class semaphores). Optionally **age** long-waiting low-priority items to avoid starvation.
- **Cancellation as scheduling.** The cheapest way to free capacity is to cancel work that no longer matters (scrolled-off images, left screen). Structured concurrency (`viewModelScope`, item-scoped jobs) makes this automatic.
- **Client rate limiting (token/leaky bucket).** Throttle a class of requests to N/sec to protect the backend and battery; honor server `429` / `Retry-After` (Ch 2).
- **Batching/coalescing.** Accumulate high-frequency small events and flush on size/time/lifecycle — turning hundreds of radio wakeups into a few.

Core insight: **treat the network as a scarce shared resource and schedule it — prioritize, bound concurrency, cancel aggressively, and batch low-value traffic.** Most of the win comes from priority + cancellation + batching, not a complex queue.

14.4 High-Level Architecture


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Components & ownership

- **`NetworkScheduler`** — admits requests by priority under a concurrency bound.
- **OkHttp `Dispatcher`** — the transport-level cap (`maxRequests`, `maxRequestsPerHost`); the scheduler sits above it for app-level priority.
- **`AnalyticsBatcher`** — coalesces background events and flushes periodically.
- **Cancellation** — item/screen-scoped jobs cancel on scroll/navigation.

Direction: sources → scheduler (priority + bound) → OkHttp → network; scroll/ nav cancels queued/in-flight low-priority work.

14.5 Detailed Component Design


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Component	Responsibility	Mechanism
`Priority`	ordered classes	enum (compareTo)
`NetworkScheduler`	admit by priority under a cap	priority-aware `Semaphore` /queue
OkHttp `Dispatcher`	transport concurrency cap	`maxRequests` / `maxRequestsPerHost`
`AnalyticsBatcher`	coalesce + flush low-value events	buffered `Channel` + timer/size
item-scoped jobs	cancel stale work	structured concurrency

Simplest effective design: a weighted/priority `Semaphore` (or per-class semaphores with reserved slots for `Immediate` / `UserVisible`) plus aggressive cancellation and analytics batching. A full priority queue is only needed when contention is high.

14.6 End-to-End Data Flow

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14.7 Complete Kotlin Implementation

Package: `com.streamcart.core.network.schedule`.

14.7.1 Priority + bounded-concurrency scheduler

```
enum class Priority { Immediate, UserVisible, Prefetch, Background }   // natural order =
priority

@Singleton
class NetworkScheduler @Inject constructor() {
    // Reserve capacity so low-priority floods can't starve user-visible work.
    private val semaphores = mapOf(
        Priority.Immediate to Semaphore(permits = 6),
        Priority.UserVisible to Semaphore(permits = 4),
        Priority.Prefetch to Semaphore(permits = 2),
        Priority.Background to Semaphore(permits = 1),
    )

    suspend fun <T> schedule(priority: Priority, block: suspend () -> T): T {
        val sem = semaphores.getValue(priority)
        return sem.withPermit { block() }     // suspends until a slot of this class is free
    }
}
```

> This per-class reservation is the pragmatic 80/20: user-visible and immediate work always have dedicated slots, so a prefetch/analytics burst can't block them. A single global priority queue is an option when you need strict global ordering (shown in §14.11).

14.7.2 OkHttp dispatcher tuning (transport cap)

```
// From Ch 1 NetworkModule — cap transport-level concurrency.
OkHttpClient.Builder()
    .dispatcher(Dispatcher().apply {
        maxRequests = 12            // total in-flight
        maxRequestsPerHost = 6      // per host (avoid hammering one backend)
    })
    .build()
```

14.7.3 Cancellation on scroll (item-scoped work)

```
// Each list item owns its load; leaving composition cancels it (structured concurrency).
@Composable
fun ProductCell(product: Product, scheduler: NetworkScheduler) {
    // Coil/image library already cancels on dispose; for custom loads:
    LaunchedEffect(product.id) {
        scheduler.schedule(Priority.UserVisible) { loadDetails(product.id) }
    }  // cancelled automatically when the cell leaves the viewport
}
```

14.7.4 Client-side rate limiting (token bucket)

```
class RateLimiter(private val permitsPerSec: Int, private val clock: () -> Long =
System::nanoTime) {
    private val mutex = Mutex()
    private var tokens = permitsPerSec.toDouble()
    private var last = clock()

    suspend fun acquire() = mutex.withLock {
        val now = clock()
        tokens = (tokens + (now - last) / 1e9 *
permitsPerSec).coerceAtMost(permitsPerSec.toDouble())
        last = now
        if (tokens < 1) {
            val waitSec = (1 - tokens) / permitsPerSec
            delay((waitSec * 1000).toLong())     // throttle; cancellable
            tokens = 0.0
        } else tokens -= 1
    }
}
```

14.7.5 Analytics batching (coalesce + flush)

```
@Singleton
class AnalyticsBatcher @Inject constructor(
    @ApplicationScope private val scope: CoroutineScope,
    private val api: AnalyticsApi,
) {
    private val buffer = Channel<Event>(capacity = Channel.UNLIMITED)

    init {
        scope.launch {
            val batch = ArrayList<Event>(MAX_BATCH)
            while (true) {
                batch += buffer.receive()                       // wait for first event
                withTimeoutOrNull(FLUSH_MS) {                   // then accumulate briefly
                    while (batch.size < MAX_BATCH) batch += buffer.receive()
                }
                runCatching { api.send(batch.toList()) }        // one request for many events
                batch.clear()
            }
        }
    }
    fun log(event: Event) { buffer.trySend(event) }             // flush also on app background
(Ch 44)
    private companion object { const val MAX_BATCH = 50; const val FLUSH_MS = 5_000L }
}
```

14.7.6 Module structure

```
:core:network/schedule  NetworkScheduler, Priority, RateLimiter
:core:analytics         AnalyticsBatcher (Ch 82)
:core:network           OkHttp Dispatcher tuning (Ch 1)
```

14.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Low-priority flood blocks UI	starvation of user-visible	per-class reserved permits
Scroll leaves stale loads running	wasted bandwidth	item-scoped cancellation (structured concurrency)
Rate limiter token math races	over/under-throttle	`Mutex`-guarded token bucket
Batch flush vs new events	lost/duplicated events	single consumer coroutine; `Channel` buffer
Cancellation vs in-flight	half-done work	coroutine cancellation propagates to OkHttp `Call` (Ch 1)
Priority inversion (low holds a slot)	high waits	reserve slots per class; optional aging

- **Structured concurrency makes cancellation the primary scheduling tool:** leaving a screen/cell cancels its jobs, freeing capacity for what's now visible — no manual queue surgery.
- **Per-class semaphores** prevent priority inversion/starvation simply; a global priority queue needs explicit aging to be fair.

14.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
429 (client too aggressive)	`RateLimited`	back off (Ch 2); lower concurrency; honor `Retry-After`	brief slowdown
Prefetch failure	error	ignore/retry later (low value)	invisible
User-visible request starved	latency	reserved permits prevent it	fast
Analytics flush fails	send error	retry batch / persist to disk (outbox, Ch 31)	no data loss
Cancellation storm (fast scroll)	many cancels	cheap; frees capacity	smooth
Background work in Doze	can't run	defer to WorkManager (Ch 30/47)	runs when allowed

 14.9 Failure Scenarios and Recovery

> **Deprioritize, don't fail:** prefetch/analytics that can't run now should wait or be dropped silently — never surface an error for work the user didn't ask for.

14.10 Scaling the Implementation

- **Level 1.** Just tune OkHttp `Dispatcher` caps + cancel on scroll (image libs do this already).
- **Level 2.** Add priority classes with per-class reserved permits; batch analytics.

- **Level 3 — production.** Client rate limiting for background classes; honor 429/`Retry-After`; lifecycle-aware flush; prefetch within budget.
- **Level 4 — large.** Adaptive concurrency (shrink on 429/poor network, grow on good); network-quality-aware prefetch (Ch 68/91); image priority tied to viewport (Ch 56).
- **Level 5 — enterprise.** A unified request scheduler with remote-configurable policies (Ch 41), per-endpoint budgets, server-coordinated client limits, and observability on queue depth/priority latency.

Avoid overengineering: most apps need only dispatcher caps + cancel-on-scroll

- analytics batching. A full priority queue with aging is justified only under heavy contention.

14.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Priority	Per-class reserved semaphores	Global priority queue	Per-class is simple, no starvation; global queue for strict ordering (needs aging)
Concurrency cap	OkHttp `Dispatcher`	App-level `Semaphore`	Dispatcher for transport; semaphore for app-level priority on top
Stale work	Cancel (structured concurrency)	Deprioritize/keep	Cancel for truly unneeded (scrolled off); deprioritize if result may still be used
Rate limiting	Token bucket	Leaky bucket / fixed window	Token bucket allows bursts; leaky smooths; fixed window simplest
Analytics	Batch + flush	Send per event	Batch for battery/network; per-event only for rare critical events
Background	In-app scheduler	WorkManager (Ch 30)	In-app for foreground prioritization; WorkManager for deferrable/guaranteed

Global priority queue vs per-class semaphores: a single global queue gives exact priority ordering but can **starve** low priority and needs aging to be fair; per-class reserved semaphores are simpler, prevent starvation of the important classes, and are usually enough. Start with per-class.

Cancel vs deprioritize: if a result is now useless (off-screen image), **cancel** to free capacity and bandwidth. If it may still be used soon, lower its priority instead so it resumes when capacity frees.

14.12 Security Considerations

- **Client rate limiting is not a security control.** A tampered client bypasses it; the **backend must rate-limit and authorize** independently (Ch 2). Client limiting is purely for UX/battery/politeness.
- **Analytics batching and PII:** batched events may carry PII; redact (Ch 28), honor consent, and don't persist sensitive events unencrypted in the batch buffer/outbox (Ch 24/31).
- **Prioritization must not leak timing:** avoid priority schemes that reveal sensitive state via request timing to a network observer.
- Client vs backend: the client schedules for UX; the backend enforces limits and access.

14.13 Performance and Resource Optimization

- **Perceived performance:** reserving slots for user-visible work is the biggest win — the content the user is looking at loads first.
- **Battery:** batching analytics and bounding concurrency cut **radio wakeups** (the radio's high-power tail state is a major drain); align background flushes with other traffic / lifecycle.
- **Cancellation** reclaims bandwidth from scrolled-away work instantly.
- **Adaptive concurrency:** reduce in-flight on `429`/poor network; increase on good — avoids both under-use and overload.
- **Profiling:** Network Inspector (concurrency, timing), battery historian / Network traffic stats, frame timing while scrolling, 429 rate.

14.14 Testing Strategy

Use virtual time and controllable blocks to assert ordering/throttling.

```
@OptIn(ExperimentalCoroutinesApi::class)
class NetworkSchedulerTest {
    @Test fun `user-visible is not blocked by background flood`() = runTest {
        val sched = NetworkScheduler()
        val order = Collections.synchronizedList(mutableListOf<String>())
        val gate = CompletableDeferred<Unit>()
        repeat(10) { launch { sched.schedule(Priority.Background) { gate.await(); order += "bg"
} } }
        launch { sched.schedule(Priority.UserVisible) { order += "vis" } }   // reserved slot
        runCurrent()
        assertTrue("vis" in order)          // ran despite background flood
        gate.complete(Unit)
    }
}

class RateLimiterTest {
    @Test fun `throttles beyond permits per second`() = runTest {
        var now = 0L
        val rl = RateLimiter(permitsPerSec = 2) { now }
        rl.acquire(); rl.acquire()           // 2 immediate
        val before = currentTime
        rl.acquire()                         // 3rd must wait ~0.5s
        assertTrue(currentTime - before >= 400)
    }
}

class AnalyticsBatcherTest {
    @Test fun `coalesces events into one request`() = runTest {
        val api = RecordingAnalyticsApi()
        val b = AnalyticsBatcher(backgroundScope, api)
        repeat(10) { b.log(Event("tap$it")) }
        advanceTimeBy(5_000)
        assertEquals(1, api.sendCount)       // one batch
        assertEquals(10, api.lastBatch.size)
    }
}
```

Also: cancellation frees a slot (a cancelled job releases its permit); 429 lowers concurrency; flush-on-background.

14.15 Production Observability

- **Metrics:** per-priority queue wait time and latency, in-flight concurrency, 429 rate, prefetch hit-rate (prefetched items actually viewed), analytics batch size, radio-active time.

- **Logs:** `net_schedule{priority, wait_ms}`, `rate_limited{class}`, `analytics_flush{count, latency_ms}`.
- **Debugging:** user-visible latency high under load → no reserved slots / unbounded background; 429 spikes → concurrency too high / no client limiting; battery drain → analytics not batched / radio thrash; wasted data → no cancel on scroll.
- **Privacy:** no PII in scheduling logs; redact batched analytics (Ch 28).

14.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Why prioritize requests?* One constrained radio; user-visible work must not queue behind prefetch/analytics.
2. *Priority classes?* Immediate / UserVisible / Prefetch / Background (ordering, not fine numbers).
3. *Why bound concurrency?* Finite radio/pool; unbounded causes contention + 429s.
4. *Cheapest way to free capacity?* Cancel work that no longer matters.
5. *Why batch analytics?* Fewer radio wakeups → battery + fewer requests.

Architecture

6. *Where does scheduling sit?* Above OkHttp — app-level priority over the transport `Dispatcher` cap.
7. *Per-class semaphores vs global queue?* Per-class simple + no starvation; global exact-order but needs aging.
8. *How tune OkHttp?* `maxRequests` / `maxRequestsPerHost` on the `Dispatcher`.

Kotlin/concurrency

9. *How cancel stale work?* Structured concurrency — item/screen-scoped jobs cancel on dispose/nav.
10. *How does cancellation reach the socket?* Coroutine cancel → OkHttp `Call` cancel (Ch 1).
11. *How throttle without blocking?* Token bucket + cancellable `delay`.

Rate limiting

12. *Token vs leaky bucket?* Token allows bursts; leaky smooths output.
13. *Is client rate limiting a security control?* No — backend must enforce.
14. *Handle 429?* Back off + honor `Retry-After` (Ch 2); lower concurrency.

Prioritization

15. *Prevent starvation?* Reserved per-class permits; optional aging.
16. *Cancel vs deprioritize?* Cancel if useless; deprioritize if maybe-used.
17. *Priority inversion?* Reserve slots for high-priority classes.

Battery/perf

18. *Why does batching save battery?* Avoids repeated radio high-power tail wakeups.
19. *Adaptive concurrency?* Shrink on 429/poor network, grow on good.

20. *Biggest perceived-perf win?* Reserved slots for user-visible content.

Scalability

21. *Remote-tune policies?* From remote config (Ch 41); per-endpoint budgets.
22. *Background deferrable work?* WorkManager (Ch 30), not the in-app scheduler.

Testing

23. *How test priority?* Flood background; assert user-visible still runs.
24. *How test rate limiting?* Fake clock; assert throttle delay.
25. *How test batching?* Log N events; assert one flush of N.

Production debugging

26. *User-visible slow under load?* No reserved slots/unbounded background.
27. *429 spikes?* Concurrency too high / no client limiting.
28. *Battery drain?* Analytics not batched; radio thrash.

Design-change challenges

1. **Image grid janky under prefetch.** Reserve visible slots; cancel off-screen; tie priority to viewport (Ch 56).
2. **Backend 429s under load.** Client rate limiting + adaptive concurrency + honor `Retry-After`.
3. **Analytics draining battery.** Batch + flush on timer/size/background.
4. **Prefetch wasting data on cellular.** Network-aware prefetch (Ch 68/91); disable on metered.
5. **Guaranteed background upload.** Move to WorkManager (Ch 30).

Mock walkthrough (abridged)

Clarify (contention? battery? priorities?) → define priority classes → per-class reserved semaphores + OkHttp caps → cancel-on-scroll → client rate limit + batching → handle 429/adaptive → testing.

14.17 Practical Exercises

1. **Beginner.** Tune OkHttp `maxRequests` / `maxRequestsPerHost` and cancel image loads on scroll. *Eval:* bounded concurrency; stale loads cancelled.
2. **Intermediate.** Add a per-class `NetworkScheduler`; prove user-visible runs under a background flood. *Eval:* no starvation.
3. **Advanced.** Add a token-bucket rate limiter + analytics batcher; prove throttling and coalescing. *Eval:* N events → 1 flush; throttle delay correct.
4. **Production debugging.** Fix a grid where prefetch blocks visible images and analytics drains battery. *Eval:* visible-first; batched analytics.
5. **System design challenge.** Add adaptive concurrency that shrinks on 429/poor network and grows on good. *Eval:* fewer 429s; good throughput when healthy.

(Solutions withheld unless requested.)

14.18 Chapter Summary & Reference Sheet

Key decisions: treat the network as a **scarce shared resource** and schedule it; use a few **priority classes** with **per-class reserved permits** (no starvation); **bound concurrency** at the OkHttp `Dispatcher` and app level; **cancel** scrolled-away/left-screen work via structured concurrency; **rate-limit and batch** low-value traffic (analytics) for battery and to avoid 429s; honor server `429` / `Retry-After`; client limiting is **not** security (backend enforces).

Concepts: priority scheduling, bounded concurrency, head-of-line blocking, cancellation-as-scheduling, token/leaky bucket, batching/coalescing, radio wakeups, adaptive concurrency.

Common mistakes: unbounded concurrency (contention/429); no reserved slots (user-visible starved); not cancelling stale loads (wasted data); per-event analytics (battery drain); treating client rate limiting as security; a global priority queue without aging (starvation).

Checklist:

- [ ] priority classes
- [ ] per-class reserved permits
- [ ] OkHttp caps
- [ ] cancel on scroll/nav
- [ ] client rate limiter
- [ ] analytics batching + lifecycle flush
- [ ] honor 429/`Retry-After`
- [ ] adaptive concurrency
- [ ] no PII in batches
- [ ] defer guaranteed work to WorkManager.

Key APIs: OkHttp `Dispatcher` (`maxRequests`/`maxRequestsPerHost`), `Semaphore.withPermit`, `Mutex`, `Channel`/`trySend`, `withTimeoutOrNull`, `delay`, structured concurrency/`viewModelScope`, coroutine cancellation.

References:

- OkHttp `Dispatcher`: https://square.github.io/okhttp/4.x/okhttp/okhttp3/-dispatcher/
- Coroutines: shared state & Semaphore/Mutex: https://kotlinlang.org/docs/shared-mutable-state-and-concurrency.html
- Optimize network/battery: https://developer.android.com/training/efficient-downloads
- WorkManager (deferrable work): https://developer.android.com/topic/libraries/architecture/workmanager
- Reduce network battery drain: https://developer.android.com/topic/performance/power/network

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; scheduler + rate limiter + batcher + OkHttp tuning + cancellation; concurrency (starvation/inversion); failure (deprioritize-not-fail); 5 levels; per-class-vs- global and cancel-vs-deprioritize trade-offs; security (not-a-control); perf (radio wakeups); virtual-time tests; observability; 28 Q&A + 5 challenges. ✅

- **Expand later:** image viewport priority (Ch 56), network-aware prefetch (Ch 68/91), WorkManager (Ch 30), analytics pipeline (Ch 82).
- **Interview coverage:** exceeds ≥25 bar. **Part II (networking patterns) core is well-covered (Ch 9–14).**